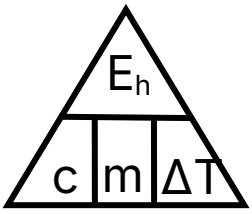
# Calculating Heat Energy, $E_h$

The formula for calculating heat (or thermal) energy is:



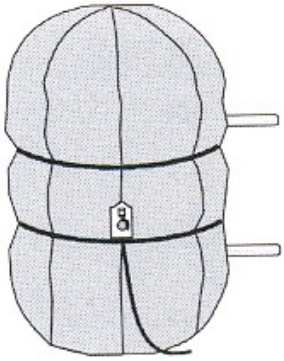

**$E_h$= cmΔT**

Heat Energy = specific heat capacity of a material x mass x change in temperature

- **Energy** is measured in **Joules** (**J**)
- **Specific heat capacity** is measured in **$Jkg^{-1}K^{-1}$** (kJ/kgK) and is the amount of energy required to raise the temperature of 1kg of the material by 1°C
- **Mass** is measured in **kg**
- **Change in temperature** (**ΔT**)('delta T') is measured in **Celsius** (**°C**) or Kelvin (**K**)

**Example**

A hot water tank contains 200 litres of water at 18°C. Calculate how much energy is required to raise the temperature of the water to 50°C. 1 litre of water has a mass of 1kg. The specific heat capacity of water is 4180 $Jkg^{-1}K^{-1}$. The specific heat capacity of a liquid changes depending on the type of liquid. The provided data booklet will specify what value to use.



**$E_h$= cmΔT**

= 200 x 4180 x 32

= 26,752,000 J

= 26.752 MJ

= **27 MJ** to two sig. fig.

What are my knowns?
m = 200kg
c = 4180 $Jkg^{-1}K^{-1}$
$\Delta T = t_2 - t_1 = 50 - 18 =$ 32°C

**Questions**

*(Calculate answer to two significant figures)*

1. Using a kettle with 1.5 Litre of water and a thermometer, work out the heat energy needed to raise the temperature from 5°C to:
   a) 50°C
   b) 40°C
   c) 80°C
   d) 100°C

2. Calculate the heat energy required to heat 1.75kg of water from a temperature of 18°C until it begins to boil.

3. 57kJ of thermal (heat) energy is supplied to 1.7kg of oil having a specific heat capacity of 2.7 kJ/kgK. If the initial temperature of the oil is 3°C, what will be its final temperature?